Configure HA Zookeeper

Zoo.cfg

Same as hdfs and yarn configuration files, Zookeeper should be configured through its `/zookeeper/conf/zoo.cfg` file

```
cp zoo_sample.cfg zoo.cfg

vi zoo.cfg
```

```
# The number of milliseconds of each tick
tickTime=2000
# The number of ticks that the initial
# synchronization phase can take
initLimit=10
# The number of ticks that can pass between
# sending a request and getting an acknowledgement
syncLimit=5
# the directory where the snapshot is stored.
# do not use /tmp for storage, /tmp here is just
# example sakes.
dataDir=/zoo_data/
# the port at which the clients will connect
clientPort=2181
# the maximum number of client connections.
# increase this if you need to handle more clients
#maxClientCnxns=60
#
# Be sure to read the maintenance section of the
# administrator guide before turning on autopurge.
#
# https://zookeeper.apache.org/doc/current/zookeeperAdmin.html#sc_maintenance
#
# The number of snapshots to retain in dataDir
#autopurge.snapRetainCount=3
# Purge task interval in hours
# Set to "0" to disable auto purge feature
#autopurge.purgeInterval=1
server.1=master1:2888:3888
server.2=master2:2888:3888
server.3=worker1:2888:3888
## Metrics Providers
#
# https://prometheus.io Metrics Exporter
#metricsProvider.className=org.apache.zookeeper.metrics.prometheus.PrometheusMetricsProvider
#metricsProvider.httpHost=0.0.0.0
#metricsProvider.httpPort=7000
#metricsProvider.exportJvmInfo=true
```

Core Timing Settings

Parameter	Value	Meaning
tickTime=2000	2000ms (2s)	The basic time unit in milliseconds. Used for heartbeats and session timeouts.
initLimit=10	10 ticks (20s)	How long follower nodes have to initially connect and sync with the leader.
syncLimit=5	5 ticks (10s)	How long a follower can be out of sync with the leader before being dropped.

Storage & Networking

Parameter	Value	Meaning
dataDir=/zoo_data/	/zoo_data/	Directory where ZooKeeper stores its snapshots (in-memory database copies).
clientPort=2181	2181	The port clients use to connect to ZooKeeper (default standard port).
#maxClientCnxns=60	60 (commented out)	Max simultaneous client connections. Commented out = unlimited.

Cluster / Ensemble Configuration

```
server.1=master1:2888:3888
server.2=master2:2888:3888
server.3=worker1:2888:3888
```

It means I have 3 servers that has the Zookeeper processes (Quorum) :

- master1 on server1
- master2 on server2

- worker1 on server3

those servers communicate with each other through 2 ports depending on purpose of communication:

1) Leader Election communication 3888

- On startup of a cluster or leader dies, an election begins
- All nodes talk to each other on port 3888 to vote for a new leader
- Each node proposes itself and compares "credentials" (transaction ID, server ID)
- The node with the most up-to-date data wins

Without this, if the leader crashes, no one can take over and the entire cluster stops working.

2) Quorum (Data) Communication

- After a leader is elected, followers connect to the leader on this port
- All write requests are forwarded to the leader through this port
- The leader broadcasts data changes back to followers for replication

Without this, nodes can't stay in sync. If the leader has data that followers don't, the cluster becomes inconsistent.

The myid File

What is it?

A tiny file placed in the dataDir (/zoo_data/) on each server containing a single number — the server's ID.

it's just indicator for the zookeeper process for which server is it on (Must match the node it's on)

why? because it has to know which server to listen on and which other servers to connect

why it cannot just detect its host name?
because the server can has multiple hostnames, Multiple network interfaces, docker/cloud/Kubernetes are not reliable

zkfc command understand

what is that mean?
`hdfs zkfc -formatZK` creates the necessary data structures inside ZooKeeper so that HDFS High Availability automatic failover can work.

`-formatZK`
It creates a special folder (ZNode) inside ZooKeeper that HDFS HA will use.

Think of ZooKeeper as a File System:

```
After running hdfs zkfc -formatZK:
──────────────────────────────────────────────────
ZooKeeper Data Tree:

/
├── zookeeper/
├── kafka/
├── hbase/
└── hadoop-ha/                    ← 🆕 CREATED!
    └── mycluster/                ← 🆕 Your HDFS cluster name
```

```
├── ActiveStandbyElectorLock  ← 🆕 For leader election
└── ActiveBreadCrumb          ← 🆕 Tracks who is Active
```

ActiveStandbyElectorLock

- This is a LOCK
- The ACTIVE NameNode's ZKFC holds this lock
- Only ONE can hold it at a time
- If lock is released → failover triggers!

ActiveBreadCrumb

- Records WHICH NameNode was last Active
- Used for fencing (preventing split-brain)

for further information: [Deep dive about Znodes]